CLASSIFICATION OF TORSION GROUPS OF Q-CURVES OVER NUMBER FIELDS OF DEGREES THREE, FIVE AND SEVEN

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Allowed standard prerequisites (editorial).

- Q-curves, elliptic torsion and Galois images under isogeny, rational-j and CM torsion classifications. Original definitions, full group lists and source-local structural/exclusion proofs with complete author bibliography retained.

1. Introduction

Let E/K be an elliptic curve, where K is a number field. By the Mordell–Weil theorem, the group $E(K)$ is finitely generated and hence of the form $T \times \mathbb{Z}^r$ for some nonnegative integer r, where T is a finite Abelian group called the *torsion subgroup* of E.

A typical question one might ask is to determine all possible torsion subgroups of elliptic curves over a given number field. For elliptic curves over $\mathbb{Q}$, we have the following famous result by Mazur.

Theorem 1.1. [14] *Let $E/\mathbb{Q}$ be an elliptic curve. Then $E(\mathbb{Q})_{tors}$ is isomorphic to one of the following* 15 *groups:*

$$\begin{aligned} &\mathbb{Z}/N_1\mathbb{Z} && \text{with } N_1 = 1, \dots, 10, 12, \\ &\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2N_2\mathbb{Z} && \text{with } N_2 = 1, \dots, 4. \end{aligned}$$

There exist infinitely many $\overline{\mathbb{Q}}$-isomorphism classes for each such torsion subgroup.

In this article, we determine all possible torsion subgroups of $\mathbb{Q}$-curves over number fields of degree 3, 5 and 7. Let us first define $\mathbb{Q}$-curves. For a field K, we will denote its absolute Galois group $\mathrm{Gal}(\overline{K}/K)$ by G_K.

Definition 1.2. We say an elliptic curve E over a number field K is a $\mathbb{Q}$-*curve* if it is $\overline{K}$-isogenous to all its $G_{\mathbb{Q}}$-conjugates.

For example, all elliptic curves isogenous to an elliptic curve with rational j-invariant are $\mathbb{Q}$-curves. Indeed, if E/K is $\overline{K}$-isogenous to an elliptic curve E' with rational j-invariant, then there is a twist E'' of E' which is defined over $\mathbb{Q}$, and an isogeny $\varphi : E \to E''$. Then there is also an isogeny $\varphi^\sigma : E^\sigma \to (E'')^\sigma$, but $(E'')^\sigma = E''$, and the composition of φ and the dual of φ^σ is an isogeny from E to E^σ.

Over number fields of odd degree, Cremona and Najman have proven that all non-CM $\mathbb{Q}$-curves arise in this way, with the isogeny even being defined over the base field of the $\mathbb{Q}$-curve.

Proposition 1.3. [4, Theorem 2.7.] *If E is a non-CM $\mathbb{Q}$-curve defined over an odd degree number field K, then E is K-isogenous to an elliptic curve E' with rational j-invariant.*

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Remark 1.4. As any isogeny is a composition of a cyclic isogeny and the multiplication–by–m map for some integer m, we may assume that the isogeny from the above proposition is cyclic.

The classification of torsion of base changes of rational elliptic curves, and more generally of elliptic curves with rational j-invariant, is known for cubic, quintic and septic fields.

Theorem 1.5. [15, Theorem 1.] *Let $E/\mathbb{Q}$ be a rational elliptic curve, and let $K/\mathbb{Q}$ be a cubic extension. Then $E(K)_{tors}$ is isomorphic to one of the following 20 groups:*

$$\begin{aligned} &\mathbb{Z}/N_1\mathbb{Z} && \text{with } N_1 = 1,\dots,10,12,13,14,18,21,\\ &\mathbb{Z}/2\mathbb{Z}\oplus\mathbb{Z}/2N_2\mathbb{Z} && \text{with } N_2 = 1,2,3,4,7. \end{aligned}$$

There exist infinitely many $\overline{\mathbb{Q}}$-isomorphism classes for each such torsion subgroup except for $\mathbb{Z}/21\mathbb{Z}$, for which there is only one elliptic curve with this torsion, which is defined over $\mathbb{Q}(\zeta_9)^+$.

Theorem 1.6. [6, Theorem 1] *Let $E/\mathbb{Q}$ be a rational elliptic curve, and let $K/\mathbb{Q}$ be a quintic extension. Then $E(K)_{tors}$ is isomorphic to one of the following 17 groups:*

$$\begin{aligned} &\mathbb{Z}/N_1\mathbb{Z} && \text{with } N_1 = 1,\dots,10,11,12,25\\ &\mathbb{Z}/2\mathbb{Z}\oplus\mathbb{Z}/2N_2\mathbb{Z} && \text{with } N_2 = 1,2,3,4. \end{aligned}$$

There exist infinitely many $\overline{\mathbb{Q}}$-isomorphism classes for each such torsion subgroup except for $\mathbb{Z}/11\mathbb{Z}$, for which there are three elliptic curves with this torsion over $\mathbb{Q}(\zeta_{11})^+$.

Theorem 1.7. [7, follows from Proposition 7.7] *Let $E/\mathbb{Q}$ be a rational elliptic curve, and let $K/\mathbb{Q}$ be a septic extension. Then $E(K)_{tors}$ is isomorphic to one of the 15 groups from Theorem 1.1.*

Remark 1.8. Gužvić proved in [9, Theorem 5.1.2] that elliptic curves with rational j-invariant over an odd prime degree number field have the same list of possible torsion subgroups as base changes of rational elliptic curves.

Over quadratic number fields, there is one exception. Namely, the group $\mathbb{Z}/13\mathbb{Z}$ appears among torsion subgroups of elliptic curves with rational j-invariant, and does not appear among torsion subgroups of rational elliptic curves. For example, consider the elliptic curve 100.1-e2 from the LMFDB.

The main results of this article are the following.

Theorem 1.9. *Let $K/\mathbb{Q}$ be a cubic extension and let E/K be a $\mathbb{Q}$-curve. Then $E(K)_{tors}$ is isomorphic to one of the 20 groups appearing in Theorem 1.5.*

Theorem 1.10. *Let $K/\mathbb{Q}$ be a quintic extension and let E/K be a $\mathbb{Q}$-curve. Then $E(K)_{tors}$ is isomorphic to one of the 17 groups appearing in Theorem 1.6.*

Theorem 1.11. *Let $K/\mathbb{Q}$ be a septic extension and let E/K be a $\mathbb{Q}$-curve. Then $E(K)_{tors}$ is isomorphic to one of the 15 groups appearing in Theorem 1.7.*

Remark 1.12. In [12], Le Fourn and Najman found all possible torsion subgroups of $\mathbb{Q}$-curves over quadratic number fields. The new groups that do not occur over the rational numbers are

$$\begin{aligned} &\mathbb{Z}/N\mathbb{Z}, && \text{for } N = 13,14,15,16,18,\\ &\mathbb{Z}/2\mathbb{Z}\oplus\mathbb{Z}/2N\mathbb{Z} && \text{for } N = 5,6,\\ &\mathbb{Z}/3\mathbb{Z}\oplus\mathbb{Z}/3N\mathbb{Z} && \text{for } N = 1,2,\\ &\mathbb{Z}/4\mathbb{Z}\oplus\mathbb{Z}/4\mathbb{Z}. \end{aligned}$$

In [4, Theorem 1.3.], Cremona and Najman have shown that when $p > 7$ is a prime and $K/\mathbb{Q}$ is an extension of degree p, every torsion subgroup $E(K)_{tors}$ of some $\mathbb{Q}$-curve E/K is one of the subgroups already appearing in Theorem 1.1.

Combining the previous remark with the results of this paper, the classification of torsion of $\mathbb{Q}$-curves over number fields of prime degree is now completed.

Corollary 1.13. *Let p be an odd prime number. If a group T appears as the torsion subgroup of some $\mathbb{Q}$-curve over a number field of degree p, then T also appears as the torsion subgroup of some elliptic curve over a number field of degree p with rational j-invariant.*

The above corollary is false for $p = 2$. However, a similar statement holds if we replace the class of $\mathbb{Q}$-curves by the class of elliptic curves which are isogenous to an elliptic curve with rational j-invariant.

Theorem 1.14. *Let p be a prime number. If a group T appears as the torsion subgroup of some elliptic curve over a number field of degree p which is $\overline{\mathbb{Q}}$-isogenous to an elliptic curve with rational j-invariant, then T also appears as the torsion subgroup of some elliptic curve over a number field of degree p with rational j-invariant.*

Remark 1.15. Under our definition, CM elliptic curves are also $\mathbb{Q}$-curves. In [3], Clark, Corn, Rice and Stankewicz classified torsion of CM elliptic curves over number fields of degree at most 13. They proved that groups which occur as torsion subgroups of CM curves over cubic fields are those from Theorem 1.1 along with $\mathbb{Z}/9\mathbb{Z}$ and $\mathbb{Z}/14\mathbb{Z}$. Over quintic fields, the only new group which appears is $\mathbb{Z}/11\mathbb{Z}$. Over septic fields, no new groups appear. In the rest of this article, we only focus on non-CM elliptic curves.

Remark 1.16. Some of the proofs in this paper were aided by the computer algebra system Magma. The auxiliary code is available in the following GitHub repository:

`https://github.com/inova3c/torsion-of-Q-curves`.

The paper is structured as follows. In Section 2, we list some general helpful results that will be used throughout the paper. In particular, we explicitly spell out the connection between the images of Galois representations of isogenous elliptic curves.

In Section 3, we prove Theorem 1.9, i.e. we solve the problem for cubic extensions. We do this case by case, eliminating each group which occurs for elliptic curves over cubic extensions and does not occur for rational elliptic curves over cubic extensions. Most of the techniques used in the cubic case are also used in the quintic and septic case, which are then discussed in Section 4. In that case, we do not have a complete list of possible torsion subgroups, but we do have finitely many cases to check as there are finitely many degrees of cyclic isogenies of elliptic curves with rational j-invariant. Finally, in Section 5, we conclude the proof of Theorem 1.14.

Acknowledgements

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2. General results

In this section, we recall some helpful results which will be used throughout the paper. Throughout the section, we will assume that all isogenies are cyclic.

It is a standard fact that every non-CM elliptic curve with rational j-invariant is a quadratic twist of an elliptic curve defined over $\mathbb{Q}$. The following well-known result describes Galois representations of quadratic twists.

Lemma 2.1. *Let E be an elliptic curve over a number field K. Let E' be a quadratic twist of E by $d \in K^\times$. Let $\psi : G_K \to \{-1, 1\}$ be the quadratic character defined by $\sigma(\sqrt{d}) = \psi(\sqrt{d})\sqrt{d}$ for $\sigma \in G_K$. Denote by $\rho_{E,N}$ and $\rho_{E',N}$ the mod N Galois representations of E and E' respectively. Then, for an appropriate choice of bases of $E[N]$ and $E'[N]$, we have*

$$\rho_{E,N}(\sigma) = \psi(\sigma)\rho_{E'N}(\sigma).$$

The proof can be found in [18, Lemma 5.1].

Since non-CM $\mathbb{Q}$-curves over odd degree number fields are isogenous to elliptic curves with rational j-invariants, it is very useful to describe what happens to the image of Galois representations after applying isogenies. We start with the following lemma which describes what happens with a point of prime order on the starting elliptic curve. We denote the cyclotomic character modulo N by χ_N.

Lemma 2.2. [2, Lemma 4.2.] *Let E_1/K be an elliptic curve with a point $P \in E_1(K)$ of prime order p and let $\varphi : E_1 \to E_2$ be an isogeny defined over K. Then either E_2 has a point of order p defined over K or for some basis of $E_2[p]$*

$$\rho_{E_2,p}(\sigma) = \begin{bmatrix} \chi_p(\sigma) & y \\ 0 & 1 \end{bmatrix}, \text{ for all } \sigma \in G_K.$$

To avoid referring to bases, we may rephrase Theorem 2.2 as follows.

Corollary 2.3. *Let E_1/F be an elliptic curve with a point $P \in E_1(K)$ of prime order p and let $\varphi : E_1 \to E_2$ be an isogeny defined over K. Then every $\alpha \in \rho_{E_2,p}(G_K)$ has eigenvalue 1.*

Combining Theorem 2.1 and Theorem 2.2, we obtain the following.

Corollary 2.4. *Let E be a $\mathbb{Q}$-curve over an odd degree number field K with a point of order p defined over K, where p is a prime number. Let $\phi : E \to E'$ be an isogeny, where E' is a curve with rational j-invariant, and let E'' be a quadratic twist of E' which is defined over $\mathbb{Q}$. Then E'' has a p-isogeny defined over K and any element of $\rho_{E'',p}(G_K)$ has eigenvalue 1 or -1.*

The following proposition, which slightly generalizes [2, Proposition 4.4.], describes what happens in the composite case.

Proposition 2.5. *Let E_1/K be an elliptic curve with a point P of order N such that $\langle P \rangle$ is fixed by G_K. Denote by $H \leq (\mathbb{Z}/N\mathbb{Z})^\times$ the subgroup $\{k \in (\mathbb{Z}/N\mathbb{Z})^\times \mid \sigma(P) = kP \text{ for some } \sigma \in G_K\}$.*

Let α be a divisor of N and $\phi : E_1 \to E_2$ the α-isogeny defined over K with kernel $\alpha \cdot P$. Then the image of the mod N representation $\rho_{E_2,N}(G_K)$ is conjugate to a subgroup of the group of all matrices of the form

$$\begin{bmatrix} h & \frac{N}{\alpha}u \\ \alpha v & z \end{bmatrix},$$

where $h \in H$.

Proof. Let $\{P, Q\}$ be a basis for $E_1[N]$. We claim that $\phi(Q)$ has order N. Indeed, if $m\phi(Q) = 0$ for $m < N$, then $mQ \in \ker \phi$. Since $\ker \phi$ is generated by a multiple of P, one has $mQ + dP = 0$ for some integer d. However, P and Q are independent, so $m = d\alpha = 0$.

Now let P_0 be a point on $E_1[N]$ such that $\frac{N}{\alpha}P_0 = P$. Let $R = \phi(P_0)$ and $S = \phi(Q)$. We claim that $\{R, S\}$ is a basis for $E_2[N]$. Both points are of order N, it remains to prove they

are independent. If $aR + bS = 0$, then $\phi(aP_0 + bQ) = 0$, so $aP_0 + bQ = d\alpha P$ for some number d. It follows that $\frac{N}{\alpha}$ divides a since aP_0 is of order N, and then we have a linear combination of P and Q being equal to 0. From here, we get $b = 0$ and the independence is proven.

We now compute the action of $\sigma \in G_K$ on both elements of this basis. We know that

$$\frac{N}{\alpha}\sigma(R) = \sigma(\phi(P)) = \phi(\sigma(P)) = h\phi(P) = h\frac{N}{\alpha}R$$

for some $h \in H$, so

$$\sigma(R) = hR + \alpha(mR + nS) = (h + m\alpha)R + n\alpha S$$

for some coefficients m, n.

Furthermore, $\sigma(S) = \sigma(\phi(Q)) = \phi(\sigma(Q))$. If we set $\sigma(Q) = uP + vQ$, we get

$$\sigma(S) = u\phi(P) + v\phi(Q) = \frac{N}{\alpha}uR + vS.$$

Note that v is the bottom right entry of $\rho_{E_1,N}(\sigma)$, and h is the top left entry of $\rho_{E_1,N}(\sigma)$. Their product hv is the determinant of $\rho_{E_1,N}(\sigma)$ which is equal to the cyclotomic character χ_N by the Weil pairing. This does not depend on the elliptic curve.

Written as a matrix, we have

$$\rho_{E_2,N}(\sigma) = \begin{bmatrix} h + m\alpha & \frac{uN}{\alpha} \\ n\alpha & v \end{bmatrix}.$$

Taking the determinant, we obtain

$$\chi_N(\sigma) = (h + m\alpha) \cdot v.$$

However, we know from before that $\chi_N(\sigma) = hv$. We get $m\alpha v = 0$ and hence $m\alpha = 0$. Thus, the top left entry of $\rho_{E_2,N}(\sigma)$ is h. □

We also spell out the special case when E_1 has a point of order N over K, not just an N-isogeny. This statement is in fact equivalent to [2, Proposition 4.4]

Corollary 2.6. *Let E_1/K be an elliptic curve with a point P of order N defined over K. Let α be a divisor of N and $\phi : E_1 \to E_2$ the α-isogeny defined over K with kernel $\alpha \cdot P$. Then the image of the mod N representation $\rho_{E_2,N}(G_K)$ is conjugate to a subgroup of the group of all matrices of the form*

$$\begin{bmatrix} 1 & \frac{N}{\alpha}u \\ \alpha v & z \end{bmatrix}.$$

The following result is also well-known, but we reprove it for the sake of completion.

Lemma 2.7. *Let K be a number field, and let M and N be positive integers. Suppose that for some elliptic curve E/K, there are generators P and Q of $E[M^2N]$ such that $[M]P$ and $[MN]Q$ lie in K. Then E is K-isogenous to an elliptic curve E_1/K such that $\rho_{E_1,M^2N}(G_K)$ is conjugate to a subgroup of the group of matrices of the form*

$$\begin{bmatrix} 1 + MN \cdot k & * \\ 0 & * \end{bmatrix}.$$

Proof. Take an M-isogeny $\phi : E \to E_1$ whose kernel is $[MN]Q$. Take a basis of the form $\{\phi(P), S\}$ of $E_1[M^2N]$. In this basis, $\rho_{E_1,M^2N}(G_K)$ has the required shape. □

We will also sometimes use the following result.

Proposition 2.8. [4, Proposition 3.3.] *Let ℓ be an odd prime. If a non-CM elliptic curve has no ℓ-isogenies defined over $\mathbb{Q}$, then it also has no ℓ-isogenies defined over K for any odd-degree number field K, unless $\ell = 7$ and $j(E) = 2268945/128$, in which case there is a 7-isogeny defined over the cubic field $\mathbb{Q}(\alpha)$, where $\alpha^3 - 5\alpha - 5 = 0$.*

Throughout the article, we will often implicitly use the following basic fact.

Lemma 2.9. *Let N be a positive integer and E/K an elliptic curve. If F/K is a finite extension of fields, then $[\rho_{E,N}(G_K) : \rho_{E,N}(G_F)]$ divides $[F : K]$.*

We remind the reader that possible torsion subgroups of CM elliptic curves are known by results from [3], so unless stated otherwise we will assume our elliptic curves are non-CM in what follows.

2.1. **Isogeny graphs.** By Theorem 1.3, any non-CM $\mathbb{Q}$-curve over an odd degree number field K is K-isogenous to an elliptic curve with rational j-invariant. In some of the proofs, we will use isogeny graphs of elliptic curves.

Let K be a number field and E/K a non-CM elliptic curve. Consider the graph whose vertices are all elliptic curves over K which are K-isogenous to E, and for each prime number p, two vertices E_1 and E_2 are connected by an edge with label p if and only if there is a p-isogeny from E_1 to E_2 which is defined over K.

This graph is called the *K-isogeny graph* of an elliptic curve E/K. We denote it by $\mathcal{G}(E/K)$. For each prime number p, the subgraph of $\mathcal{G}(E/K)$ induced by vertices which are p^k-isogenous to E for some $k \geq 0$ is called the p-primary component $\mathcal{G}_p(E/K)$ of $\mathcal{G}(E/K)$. For non-CM elliptic curves, $\mathcal{G}_p(E/K)$ is a tree, for a cycle in the graph would imply the existence of an endomorphism which is not a multiplication-by-m map.

It is well-known that for an odd prime p, any elliptic curve E over a number field K has either $0, 1, 2$ or $p + 1$ isogenies of degree p defined over K. For $p = 2$, E has either 0, 1 or 3 isogenies of degree 2. Furthermore, the following is true.

Lemma 2.10. *Let p be a prime number. Let E/K be an elliptic curve over a number field K with $p+1$ isogenies of degree p defined over K. Then K contains the unique quadratic subfield of the cyclotomic field $\mathbb{Q}(\zeta_p)$.*

Proof. If E/K has all $p + 1$ isogenies of degree p defined over K, then $\rho_{E,p}(G_K)$ consists only of scalar matrices, since the action of the Galois group fixes every subgroup of order p of $E[p] \cong (\mathbb{Z}/p\mathbb{Z})^2$. However, the determinant of a scalar matrix is a square. From the Weil pairing, it follows that the unique quadratic subfield of $\mathbb{Q}(\zeta_p)$ is contained inside K. □

Corollary 2.11. *If p is an odd prime, $K/\mathbb{Q}$ is a number field of odd degree and E/K is an elliptic curve, the p-primary component $\mathcal{G}_p(E/K)$ is a line graph.*

Proof. The graph $\mathcal{G}_p(E/K)$ is a tree and hence connected, and Theorem 2.10 implies that any vertex has degree $0, 1$ or 2. The claim follows. □

3. Torsion over cubic number fields

The list of all possible torsion subgroups of eliptic curves over cubic fields is known.

Theorem 3.1. [5, Theorem A] *Let $K/\mathbb{Q}$ be a cubic extension and E/K be an elliptic curve. Then $E(K)_{tors}$ is isomorphic to one of the following 26 groups:*

$$\mathbb{Z}/N_1\mathbb{Z} \qquad \text{with } N_1 = 1, \ldots, 16, 18, 20, 21,$$
$$\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2N_2\mathbb{Z} \qquad \text{with } N_2 = 1, \ldots, 7.$$

There exist infinitely many $\overline{\mathbb{Q}}$-isomorphism classes for each such torsion subgroup except for $\mathbb{Z}/21\mathbb{Z}$, for which there is only one elliptic curve with this torsion, which is defined over $\mathbb{Q}(\zeta_9)^+$.

To prove Theorem 1.9, it remains to prove that none of the following six groups appears as torsion of a $\mathbb{Q}$-curve over a cubic field.

$$\begin{aligned} &\mathbb{Z}/N_1\mathbb{Z} && \text{with } N_1 = 11, 15, 16, 20 \\ &\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2N_2\mathbb{Z} && \text{with } N_2 = 5, 6. \end{aligned}$$

3.1. Eliminating $\mathbb{Z}/11\mathbb{Z}$.

Proposition 3.2. *The group $\mathbb{Z}/11\mathbb{Z}$ does not occur as a torsion subgroup of a $\mathbb{Q}$-curve over a cubic number field.*

Proof. Suppose that some $\mathbb{Q}$-curve E over a cubic field K has a point P of order 11. We have the following situation:

$$E/K \xrightarrow{\phi} E'/K \xrightarrow{\sim} E''/\mathbb{Q}.$$

Here E is our starting $\mathbb{Q}$-curve, ϕ is a cyclic isogeny defined over K, E'/K is an eliptic curve with rational j-invariant, E'' is an elliptic curve defined over $\mathbb{Q}$ and $\sim$ is an isomorphism from E' to E'', i.e. E'' is a quadratic twist of E'.

By Corollary 2.4, we know that E'' has an 11-isogeny over K. However, by Proposition 2.8, E'' also has an 11-isogeny over $\mathbb{Q}$.

There are two rational j-invariants for which the corresponding elliptic curve $E/\mathbb{Q}$ has an 11-isogeny over $\mathbb{Q}$. For a fixed j-invariant, any two rational elliptic curves with this j-invariant are quadratic twists. This means that we can pick any rational elliptic curve with this j-invariant and check that its Galois image does not have the properties it would need to have by Corollary 2.4.

The j-invariants for which there is an 11-isogeny over $\mathbb{Q}$ are -11^2 and $-11 \cdot 131^3$. Examples of elliptic curves with these j-invariants with LMFDB labels `1089.c1` and `1089.c2`.

For E'' being any of these curves, the image of the mod 11 Galois representation $\rho_{E'',11}(G_{\mathbb{Q}})$ has 220 elements. Since $3 \nmid 220$ and $[\rho_{E'',11}(G_{\mathbb{Q}}) : \rho_{E'',11}(G_K)]$ divides 3, we must have $\rho_{E'',11}(G_K) = \rho_{E'',11}(G_{\mathbb{Q}})$.

Directly inspecting the images $\rho_{E'',11}(G_{\mathbb{Q}})$, there are matrices which have neither 1 nor -1 as an eigenvalue, so we reach a contradiction with Corollary 2.4. □

3.2. Eliminating $\mathbb{Z}/15\mathbb{Z}$.

Proposition 3.3. *The group $\mathbb{Z}/15\mathbb{Z}$ does not occur as a torsion subgroup of a $\mathbb{Q}$-curve over a cubic number field.*

Proof. Suppose that some $\mathbb{Q}$-curve E over a cubic field K has a point P of order 15. We have the following situation:

$$E/K \xrightarrow{\phi} E'/K \xrightarrow{\sim} E''/\mathbb{Q}.$$

Here E' is an elliptic curve over K with rational j-invariant, and E'' is a quadratic twist of E' which is defined over $\mathbb{Q}$.

Then, similarly as in the proof of Proposition 3.2, we can conclude that E'' has a 3-isogeny and a 5-isogeny over $\mathbb{Q}$, so it must have a 15-isogeny over $\mathbb{Q}$. There are four possible j-invariants for which this happens.

As in the proof of Proposition 3.2, we can take one specific elliptic curve for each of the j-invariants.

We will look at curves from the isogeny class `1600.i` on LMFDB. Note that $-I$ is contained in the mod 15 image.

By Theorem 2.1, we have $\rho_{E',15} \sim \psi \cdot \rho_{E'',15}$, where ψ is some quadratic character. Furthermore, any element of $\rho_{E',15}(G_K)$ has 1 as an eigenvalue when reduced modulo 3 and modulo 5. This means that any element of $\rho_{E'',15}(G_K)$ either has 1 as an eigenvalue modulo 3 and modulo 5, or it has -1 as an eigenvalue modulo 3 and modulo 5. Now, for each of the four curves E'' from the class 1600.i, we do the following:

- List all index 3 subgroups H of $\rho_{E'',15}(G_{\mathbb{Q}})$.
- In each of the subgroups H from the list, find a matrix A such that at least one matrix among $A \pmod 3$, $A \pmod 5$ doesn't have 1 as an eigenvalue, and at least one matrix among $A \pmod 3$, $A \pmod 5$ doesn't have -1 as an eigenvalue.

The existence a matrix $A \in H$ with this property implies that H cannot be the image $\rho_{E'',15}(G_K)$.

(1) **Case 1600.i1** There are three index 3 subgroups, so we have three cases. They are listed in order in which they appear in the auxiliary Magma program.
   (a) The matrices $\left[\begin{smallmatrix} 2 & 5 \\ 0 & 1 \end{smallmatrix}\right]$ and $\left[\begin{smallmatrix} 1 & 10 \\ 0 & 11 \end{smallmatrix}\right]$ are in the image $\rho_{E'',15}(G_K)$. They both have to be in $\rho_{E',15}(G_K)$ as their negatives don't have 1 as an eigenvalue mod 5. However, their product doesn't have 1 as an eigenvalue mod 3.
   (b) The matrices $\left[\begin{smallmatrix} 12 & 1 \\ 10 & 6 \end{smallmatrix}\right]$ and $\left[\begin{smallmatrix} 6 & 5 \\ 5 & 6 \end{smallmatrix}\right]$ are in $\rho_{E'',15}(G_K)$. They don't have -1 as eigenvalues mod 5, so they themselves have to be in $\rho_{E',15}(G_K)$. However, their product equals $-I$ mod 3, which doesn't have 1 as an eigenvalue mod 3, so we reach a contradiction.
   (c) Similarly as in the previous case, we end up with $-I$ in $\rho_{E',15}(G_K)$, which is a contradiction.

(2) **Case 1600.i2** Each of the index 3 subgroups contains at least one among $\left[\begin{smallmatrix} 2 & 0 \\ 0 & 11 \end{smallmatrix}\right]$ and $\left[\begin{smallmatrix} 2 & 9 \\ 0 & 11 \end{smallmatrix}\right]$.

(3) **Case 1600.i3** Each of the index 3 subgroups contains a matrix of the form $\left[\begin{smallmatrix} 11 & * \\ 0 & 2 \end{smallmatrix}\right]$.

(4) **Case 1600.i4** The matrix $\left[\begin{smallmatrix} 11 & 0 \\ 0 & 11 \end{smallmatrix}\right]$ is in all index 3 subgroups.

□

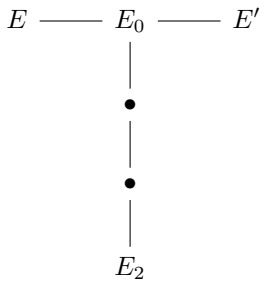
3.3. **Eliminating $\mathbb{Z}/16\mathbb{Z}$.**

Proposition 3.4. *The group $\mathbb{Z}/16\mathbb{Z}$ does not occur as a torsion subgroup of a $\mathbb{Q}$-curve over a cubic number field.*

Proof. Suppose that some $\mathbb{Q}$-curve E over a cubic field K has a point P of order 16. Then $j(E)$ is not rational.

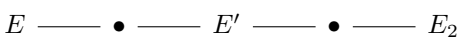
We know that E is K-isogenous to an elliptic curve E' with rational j-invariant. Since composing by an isogeny of odd degree does not effect 16-torsion, we may assume that the isogeny $\phi : E \to E'$ has degree 2^k for some positive integer k. Then $k \leq 4$ by [16, Theorem 1.2]. Suppose that $k > 2$. Consider a quadratic twist E'' of E' such that E'' is defined over $\mathbb{Q}$. Then E'' is 2^k-isogenous to a quadratic twist E''' of E. However, by [16, Proposition 3.2], it follows that for $k > 2$, any 2^k-isogeny of a rational elliptic curve which is defined over a cubic extension is in fact defined over $\mathbb{Q}$. In particular, E''' is also defined over $\mathbb{Q}$, so $j(E''') = j(E)$ is rational, a contradiction.

Thus, $k \leq 2$. Let P be a K-rational point of order 16 on E. We claim that the K-rational isogeny $\psi : E \to E_2$ with kernel $\langle P \rangle$ factors through the 2^k-isogeny $\phi : E \to E'$. Suppose for the sake of contradiction that it does not. If $k = 1$, then the dual isogeny $\hat{\phi} : E' \to E$ composed with $\psi : E \to E_2$ yields a cyclic 32-isogeny of E', a contradiction with [16, Theorem 1.2]. If $k = 2$ and the isogenies ϕ and ψ are independent, then E' has a $4 \cdot 16$-isogeny, a contradiction. If $k = 2$ and ϕ and ψ are not independent, then $\ker \phi \cap \ker \psi$ has size 2. We have the following diagram, where each of the edges denotes a 2-isogeny.



Since E' and E_2 are 16-isogenous, it follows using the same argument as before that $j(E_2)$ is rational. However, then all the elliptic curves on the path have rational j-invariant. Consider a rational quadratic twist E_0' of E_0. Then E_0' has two rational 2-isogenies, so the third of its 2-isogenies must also be rational. This implies that some twist of E is a rational elliptic curve and hence E has rational j-invariant, a contradiction. Thus, $\psi : E \to E_2$ factors through $\phi : E \to E'$.

If $k = 1$, then there is an 8-isogeny $E' \to E_2$, so E_2 has rational j-invariant. However, then $E_2 \to E$ is a 16-isogeny, so E has rational j-invariant, a contradiction. Thus, $k = 2$ and we have the following diagram.



By Theorem 2.6, it follows that $\rho_{E',16}(G_K)$ is conjugate to a subgroup of the group of all matrices of the form $\left[\begin{smallmatrix} 1 & 4x \\ 4y & z \end{smallmatrix}\right]$. Let E'' be a quadratic twist of E' which is defined over $\mathbb{Q}$. Then $\rho_{E',16}(G_K)$ is conjugate to a subgroup of the group of all matrices of the form $\left[\begin{smallmatrix} \pm 1 & 4x \\ 4y & z \end{smallmatrix}\right]$.

We see that E'' has a 4-isogeny over K which is not defined over $\mathbb{Q}$. By [16, Proposition 3.2], it follows that the image of the 2-adic Galois representation $\rho_{E'',2^\infty}(G_\mathbb{Q})$ is conjugate to H_{20}, where H_{20} is the corresponding group from the list in [17]. The group $\rho_{E'',2^\infty}(G_K)$ is its index 3 subgroup (it cannot be equal to it, as there would be no 2-isogenies over K).

However, H_{20} is defined modulo 4. This implies that $\rho_{E'',4}(G_K)$ is conjugate to an index 3 subgroup of the reduction modulo 4 of H_{20}, and hence $\rho_{E'',2^\infty}(G_K)$ is also defined modulo 4. But then $\rho_{E'',16}(G_K)$ contains all matrices of the form $I \pmod 4$, which is a contradiction since the matrix $\left[\begin{smallmatrix} 5 & 0 \\ 0 & 5 \end{smallmatrix}\right]$ is not conjugate to a matrix of the form $\left[\begin{smallmatrix} \pm 1 & 4x \\ 4y & z \end{smallmatrix}\right]$. □

3.4. Eliminating $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/10\mathbb{Z}$ and $\mathbb{Z}/20\mathbb{Z}$.

Proposition 3.5. *The groups $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/10\mathbb{Z}$ does not occur as a torsion subgroup of $\mathbb{Q}$-curves over cubic number fields.*

Proof. Suppose that $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/10\mathbb{Z}$ does occur for some $\mathbb{Q}$-curve E over a number field K. We first use Theorem 2.7 with $M = 2$, $N = 5$ to conclude that there is a 2-isogenous curve E_1 such that $\rho_{E_1,20}(G_K)$ is conjugate to a subgroup of the group of all matrices of the form $\left[\begin{smallmatrix} h & * \\ 0 & * \end{smallmatrix}\right]$, where $h \in \{1, 11\}$. We see that E_1 has a 20-isogeny $\phi : E_1 \to E_2$ and hence does not

have rational j-invariant by [16, Theorem 1.2]. The curve E_1 is also K-isogenous to an elliptic curve E'/K with a rational j-invariant. Let $\psi : E_1 \to E'$ denote this isogeny. As before, we may assume ψ is cyclic of degree $2^a \cdot 5^b$ for some $a, b \geq 0$.

Since E' has a 5-isogeny over K, it does not have a 4-isogeny. Hence the degree of $\psi : E_1 \to E'$ is divisible by 2 and is not divisible by 4, so it equals 2 or 10. It cannot be 50 by [16, Theorem 1.2].

Suppose that the degree is 10. We have the following diagram, with the labels above edges being the degrees.

$$E_1 \overset{2}{\text{——}} E_3 \overset{5}{\text{——}} E'$$

We claim that E_3 also has rational j-invariant. Consider a quadratic twist E'' of E' which is defined over $\mathbb{Q}$. By Theorem 2.8, E'' has a rational 5-isogeny. If this 5-isogeny is not to a twist of E_3, then this 5-isogenous rational elliptic curve has a 50-isogeny over K to a twist of E_1, a contradiction. Thus, E_3 is a twist of a rational elliptic curve and hence has rational j-invariant.

Hence, we may assume that the degree of ψ is 2. If ψ does not factor through ϕ, then ψ and ϕ are independent and we would obtain a 40-isogeny of E', a contradiction. Thus, ϕ factors through ψ. However, then E' is isomorphic to the starting curve E, since the 20-isogeny $E_1 \to E_2$ factors through both $E_1 \to E$ and $E_1 \to E'$, and they are both of degree 2. But then E has rational j-invariant, a contradiction. □

Proposition 3.6. *The group $\mathbb{Z}/20\mathbb{Z}$ does not occur as a torsion subgroup of a $\mathbb{Q}$-curve over a cubic number field.*

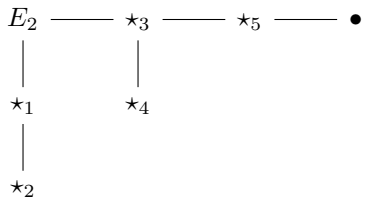
Proof. Suppose that the group $\mathbb{Z}/20\mathbb{Z}$ does occur as a torsion subgroup of a $\mathbb{Q}$-curve E/K, where K is a cubic field. Then there exists a 2-isogenous elliptic curve over K with torsion $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/10\mathbb{Z}$, and we already proved that this is impossible. □

3.5. Eliminating $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/12\mathbb{Z}$.

Proposition 3.7. *The group $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/12\mathbb{Z}$ does not occur as a torsion subgroup of a $\mathbb{Q}$-curve over a cubic number field.*

Proof. Suppose that this group does occur for some $\mathbb{Q}$-curve E over a number field K. We first use Theorem 2.7 with $M = 2$, $N = 6$ to conclude that there is a 2-isogenous curve E_1 such that $\rho_{E_1,24}(G_K)$ consists (in a suitable basis) of matrices of the form $\left[\begin{smallmatrix} h & * \\ 0 & * \end{smallmatrix}\right]$, where $h \in \{1, 13\}$. In particular, E_1 has a K-rational 24-isogeny. The curve E_1 is also K-isogenous to an elliptic curve E' with a rational j-invariant. As before, we may assume that this isogeny is cyclic of degree $2^a \cdot 3^b$ for $a, b \geq 0$. Let us factor this isogeny as $E_1 \to E_2$ of degree 3^b and $E_2 \to E'$ of degree 2^a. If E_2 has rational j-invariant, then E_2 has a 3-isogeny and an 8-isogeny over K, so it has a 24-isogeny as well. However, this is not a valid isogeny degree over cubic fields by [16, Theorem 1.2]. Thus, E_2 does not have rational j-invariant and hence $a > 0$. If $a > 2$, then E' has an 8-isogeny and a 3-isogeny, which is again impossible for the same reason. We know that E_2 has an 8-isogeny coming from the 8-isogeny of E_1. Consider the path in the 2-isogeny graph formed by these isogenies. Since the distance between E' and E_2 in the graph is 1 or 2, the position of E_2 with respect to this path is among those marked by $\star$ in the following

diagram (note that the diagram does not show the entire 2-isogeny graph).



If E' is in the position of $\star_1$, $\star_2$ or $\star_4$, it has an 8-isogeny, a contradiction. In other cases, E' has a 4-isogeny to an elliptic curve with non-rational j-invariant and an independent 2-isogeny, so $\rho_{E',2}(G_K)$ is trivial. Since E' has a 4-isogeny to an elliptic curve with non-rational j-invariant, it follows from [16, Proposition 3.2] that some quadratic twist E'' of E' which is defined over $\mathbb{Q}$ has 2-adic Galois image $\rho_{E'',2^\infty}(G_\mathbb{Q})$ equal to the group H_{20} from the database of Rouse and Zureick–Brown in [17]. Reducing modulo 4, this is a group of order 12 generated by $\left[\begin{smallmatrix}1&1\\3&0\end{smallmatrix}\right]$ and $\left[\begin{smallmatrix}3&3\\0&1\end{smallmatrix}\right]$. The group $\rho_{E'',4}(G_K)$ is a subgroup of order 4 which is contained inside the group of uppertriangular matrices. There is only one such subgroup. However, if this subgroup is equal to $\rho_{E'',4}(G_K)$, then $\rho_{E'',2}(G_K)$ is not trivial, because $\rho_{E'',4}(G_K)$ contains $\left[\begin{smallmatrix}3&3\\0&1\end{smallmatrix}\right]$. This is a contradiction since then E'' (and hence also E') does not have all 2-isogenies defined over K. □

4. Torsion over number fields of degree 5 or 7

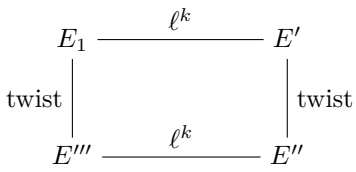
4.1. Possible prime power orders of points.

Lemma 4.1. *Let E be a $\mathbb{Q}$-curve over a number field K of prime degree $p \geq 5$. Let $\ell \neq p$ be an odd prime. Suppose that E has a point of order $\ell \neq p$ defined over K.*

Then there exists an elliptic curve E_1 which is K-isogenous to E such that E_1 has rational j-invariant and a point of order ℓ.

Proof. Let E be a $\mathbb{Q}$-curve over K. Let E' be the K-isogenous elliptic curve with rational j-invariant. We can factor the isogeny $E \to E'$ by writing it down as $E \to E_1 \to E'$, where the isogeny $E \to E_1$ has degree coprime to ℓ and the isogeny $E_1 \to E'$ has order ℓ^k for some $k \geq 0$. Then E_1 is a $\mathbb{Q}$-curve isogenous to E which also has a point of order ℓ over K.

If $\ell^k = 1$, then E_1 is isomorphic to E and hence has rational j-invariant. Otherwise, let E'' be the quadratic twist of E' which is defined over $\mathbb{Q}$. Furthermore, let E''' be the corresponding quadratic twist of E_1, such that we have the following diagram (the vertical lines are quadratic twists, and the horizontal lines are ℓ^k-isogenies).



Note that p does not divide $\ell \cdot (\ell - 1)$ unless $p = 5$ and $\ell = 11$, so in those cases $\rho_{E'',\ell^k}(G_\mathbb{Q}) = \rho_{E'',\ell^k}(G_K)$ and it follows that any ℓ^k-isogeny of E'' defined over K is in fact already defined over $\mathbb{Q}$.

Now suppose that $\ell = 11$ and $p = 5$. Then E'' is a rational elliptic curve with an 11-isogeny over a quintic extension, so by Theorem 2.8 it must have a rational 11-isogeny. However,

then $j(E'') = j(E')$ belongs to the set $\{-11^2, -11 \cdot 131\}$. The elliptic curves with these j-invariants have only one 11-isogeny (the rational one) over quintic extensions of $\mathbb{Q}$. They have no isogenies of odd prime order over $\mathbb{Q}$, so their isogeny class does not increase when taking quintic extensions. It follows that they are the only $\mathbb{Q}$-curves over quintic extensions with a point of order 11. □

The above result implies that any possible prime order of a point already appears among elliptic curves with rational j-invariant.

In particular, by Theorem 1.6 there are no points of prime order greater than 11 over degree 5 extensions, and by [15, Proposition 7.7], there are no points of prime order greater than 7 over degree 7 extensions. Since the points of order 11 have just been discussed, we may now only focus on powers of $2, 3, 5, 7$.

Lemma 4.2. *A $\mathbb{Q}$-curve over a number field of degree 5 or 7 does not have a point of order 16 or 27.*

Proof. Note that 5 and 7 do not divide the orders of $\mathrm{GL}_2(\mathbb{Z}/2\mathbb{Z})$ and $\mathrm{GL}_2(\mathbb{Z}/3\mathbb{Z})$.

Suppose that there is a $\mathbb{Q}$-curve E over a degree 5 or 7 extension with a 16-isogeny or a 27-isogeny and let E' be an isogenous curve with rational j-invariant, E'' the quadratic twist of E' which is rational and E''' the corresponding twist of E, as in the diagram in the proof of Theorem 4.1. Without loss of generality we may assume that the isogeny $E \to E'$ is a 2^a3^b-isogeny for some $a, b \geq 0$.

However, the corresponding 2^a3^b-isogeny $E'' \to E'''$ is defined over $\mathbb{Q}$ since 5 and 7 do not divide $\mathrm{GL}_2(\mathbb{Z}/2\mathbb{Z})$ or $\mathrm{GL}_2(Z/3\mathbb{Z})$, so E''' is a rational elliptic curve and its twist E has rational j-invariant. The claim now follows by Theorem 1.6 and Theorem 1.7. □

Lemma 4.3. *If $E/\mathbb{Q}$ is an elliptic curve with a 5-isogeny, the index of the image of its 5-adic Galois representation in $\mathrm{GL}_2(\mathbb{Z}_5)$ is not divisible by 25.*

Proof. This follows immediately from [8, Theorem 2.], since the index of a Sylow pro-5 subgroup of $\mathrm{GL}_2(\mathbb{Z}_5)$ is not divisible by 25. □

Lemma 4.4. *There are no $\mathbb{Q}$-curves over degree 5 number fields with a point of order 125, and no $\mathbb{Q}$-curves over degree 7 number fields with a point of order 25.*

Proof. Suppose that E is a $\mathbb{Q}$-curve defined over a quintic extension of $\mathbb{Q}$ with a point of order 125. We know that E is K-isogenous to an elliptic curve with rational j-invariant E'. Without loss of generality, we may assume that the isogeny degree is 5^a for some $a \geq 1$. If $a > 3$, then E' has a 625-isogeny over a quintic extension, which is impossible by [16, Theorem 1.3]. Thus $a \leq 3$. The elliptic curve E also has a 125-isogeny defined over K whose kernel is the point of order 125 on E. Since the 5-isogeny subgraph is a line by Theorem 2.11, it follows that the isogeny $E \to E'$ either factors through the 125-isogeny whose kernel is the torsion point or the isogenies are independent. If they are independent, then E' has a 5^{3+a}-isogeny, which is again a contradiction. Thus, $E \to E'$ factors through the 125-isogeny. Then, by Theorem 2.6, $\rho_{E',125}(G_K)$ is conjugate to a subgroup of the group of all matrices of the form

$$\begin{bmatrix} 1 & 5^a \cdot x \\ 5^{3-a} \cdot y & * \end{bmatrix}$$

for some $a \in \{0, 1, 2, 3\}$.

If E'' is a quadratic twist of E' which is a rational elliptic curve, then $\rho_{E'',125}(G_K)$ is conjugate to a subgroup of the group of all matrices of the form

$$\begin{bmatrix} \pm 1 & 5^a \cdot x \\ 5^{3-a} \cdot y & * \end{bmatrix}$$

for some $a \in \{0, 1, 2, 3\}$.

This means that $\rho_{E'',125}(G_K)$ has at most $2 \cdot 25 \cdot 4 \cdot 125 = 2^3 \cdot 5^5$ elements. It follows that the index of $\rho_{E'',125}(G_K)$ is divisible by $\#\mathrm{GL}_2(\mathbb{Z}/125\mathbb{Z})/(2^3 \cdot 5^5) = 2^2 \cdot 3 \cdot 5^4$, so it is divisible by 5^4. It follows that the index of $\rho_{E'',125}(G_{\mathbb{Q}})$ is divisible by 5^3. Since E'' has a 5-isogeny over $\mathbb{Q}$, Theorem 4.3 implies that $\rho_{E'',5^\infty}(G_{\mathbb{Q}})$ is not divisible by 25, so we reach a contradiction.

Now suppose that E is defined over a degree 7 extension K and has a point of order 25. Let E' be the isogenous curve which has rational j-invariant and let 5^a be the isogeny degree. We have $a \leq 2$ since 125 is not a possible isogeny degree of an elliptic curve with rational j-invariant.

The 5-isogeny graph is a line so we can again conclude that the isogeny $E \to E'$ either factors through the isogeny whose kernel is a point of order 25 or they are independent. If they are independent, then E' has a 125-isogeny over an extension of degree 7, a contradiction, so $E \to E'$ factors through the isogeny whose kernel is the torsion point. If we let E'' be the quadratic twist of E' which is defined over $\mathbb{Q}$, we know by Theorem 2.6 that $\rho_{E'',25}(G_K)$ is contained in the subgroup of the group of all matrices of the form

$$\begin{bmatrix} \pm 1 & 5^a \cdot x \\ 5^{2-a} \cdot y & * \end{bmatrix}$$

for some $a \in \{0, 1, 2\}$. Since $7 \nmid |\mathrm{GL}_2(\mathbb{Z}/5\mathbb{Z})|$, it follows that $\rho_{E'',25}(G_K) = \rho_{E'',25}(G_{\mathbb{Q}})$. We can now compute that the index of $\rho_{E'',25}(G_{\mathbb{Q}})$ is at least $3 \cdot 2^2 \cdot 5^2$, which is again divisible by 25, a contradiction with Theorem 4.3. □

Lemma 4.5. *There are no $\mathbb{Q}$-curves over number fields of degree 5 or 7 with a point of order 49.*

Proof. Let E be a $\mathbb{Q}$-curve over a degree 5 extension K with a point of order 49, E' be the K-isogenous curve with rational j-invariant and E'' the twist of E' which is rational. We may assume that the isogeny $E \to E'$ has degree 7^a for $a \geq 1$. Since 7^2 is not a possible isogeny degree of elliptic curves with rational j-invariant over quintic fields, it follows that $a = 1$ and that the isogeny $E \to E'$ factors through the isogeny whose kernel is the point of order 49. By Theorem 2.5, E' has two independent 7-isogenies over K. Then E'' also has two independent 7-isogenies. Since $5 \nmid |\mathrm{GL}_2(\mathbb{Z}/7\mathbb{Z})|$, they are also defined over $\mathbb{Q}$. However, there then exists a $\mathbb{Q}$-isogenous elliptic curve with a 49-isogeny over $\mathbb{Q}$, which is a contradiction.

Now suppose that E is a $\mathbb{Q}$-curve over a degree 7 extension K with a point of order 49, E' the K-isogenous curve with rational j-invariant and E'' the twist of E' which is rational. We may assume that the isogeny $E \to E'$ has degree 7^a for $a \geq 1$. Since 7^3 is not a possible isogeny degree of an elliptic curve with rational j-invariant over a septic field, it follows that the isogeny $E \to E'$ factors through the isogeny whose kernel is the point of order 49. Furthermore, since a twist E'' of E' has a rational 7-isogeny, we may assume that we have the following diagram.

$$E \text{ —— } E' \text{ —— } E_2$$

Here E' and E_2 both have rational j-invariant.

by Theorem 2.5, it follows that for the quadratic twist E'' of E' the group $\rho_{E'',49}(G_K)$ is a subgroup of the group of all matrices of the form

$$\begin{bmatrix} \pm 1 & 7\cdot x \\ 7\cdot y & * \end{bmatrix}$$

Since E'' has a 7-isogeny over $\mathbb{Q}$, [13, Theorem 3.9] implies that that the 7-adic representation of E'' over $\mathbb{Q}$ is defined modulo 7, i.e. it contains all matrices congruent to I modulo 7.

To summarize, $\rho_{E'',49}(G_{\mathbb{Q}})$, once a suitable basis has been chosen, has a subgroup of index 7 contained in the group of all matrices of the form $\left[\begin{smallmatrix} \pm 1 & 7\cdot x \\ 7\cdot y & * \end{smallmatrix}\right]$, which has order $7^3 \cdot 2 \cdot 6$ and $\rho_{E'',49}(G_{\mathbb{Q}})$ contains the group of all matrices congruent to I modulo 7, which has order 7^4. This means that the order of $\rho_{E'',49}(G_K)$ must be divisible by 7^3.

Without loss of generality, taking another quadratic twist if needed, we may assume that $\rho_{E'',49}(G_K)$ also contains $-I$. We know that $\rho_{E''.49}(G_K)$ has surjective determinant in $(\mathbb{Z}/7\mathbb{Z})^\times$. The preimages under $\det : \rho_{E'',7}(G_K) \to (\mathbb{Z}/7\mathbb{Z})^\times$ of all elements of $(\mathbb{Z}/7\mathbb{Z})\times$ have the same number of elements, and this number is divisible by 2 since $\det(-I) = 1$. It follows that $\rho_{E'',7}(G_K)$ and consequently $\rho_{E'',49}(G_K)$, has order divisible by 12. But then $\#\rho_{E'',49}(G_K)$ is divisible by $7^3 \cdot 12$, and $7 \cdot \#\rho_{E'',49}(G_K) = \#\rho_{E'',49}(G_{\mathbb{Q}})$ divides $7^4 \cdot 12$. It follows that we must have equality everywhere, so $\rho_{E'',49}(G_K)$ is *equal* to the group of all matrices of the form $\left[\begin{smallmatrix} \pm 1 & 7\cdot x \\ 7\cdot y & * \end{smallmatrix}\right]$. Since $\rho_{E'',49}(G_{\mathbb{Q}})$ contains all matrices of this form and all matrices congruent to I (mod 7), it follows that $\rho_{E'',49}(G_{\mathbb{Q}})$ is the group of all matrices of the form $\left[\begin{smallmatrix} \pm 1+7\cdot z & 7\cdot x \\ 7\cdot y & * \end{smallmatrix}\right]$. But then E'' has two independent rational 7-isogenies, a contradiction. □

4.2. **Possible odd orders of points.**

Lemma 4.6. *There are no $\mathbb{Q}$-curves over number fields of degree 5 or 7 with a point of order 35.*

Proof. If E is such a $\mathbb{Q}$-curve, then there exists an elliptic curve with rational j-invariant in the isogeny class with a 35-isogeny over an extension of degree 5 or 7, which is impossible by [16, Theorem 1.3]. □

Lemma 4.7. *There are no $\mathbb{Q}$-curves over number fields of degree 5 or 7 with a point of order 15 or 21.*

Proof. Suppose that there is a $\mathbb{Q}$-curve E over a number field K of degree 5 or 7 with a point of order 15 over an extension of degree 5 or 7. Then an elliptic curve with rational j-invariant in the isogeny class has a quadratic twist which has a rational 15-isogeny or a 21-isogeny. We can now copy the proof of Theorem 3.3 to conclude that we must do the following for each of the four curves E'' from the class 1600.i:

- List all index 5 or 7 subgroups H of $\rho_{E'',15}(G_{\mathbb{Q}})$.
- In each of the subgroups H from the list, find a matrix A such that at least one matrix among A (mod 3), A (mod 5) doesn't have 1 as an eigenvalue, and at least one matrix among A (mod 3), A (mod 5) doesn't have -1 as an eigenvalue.

The existence of a matrix $A \in H$ with this property implies that H cannot be the image $\rho_{E'',15}(G_K)$.

In fact, there are no index 7 subgroups as 7 doesn't divide the order of $\mathrm{GL}_2(\mathbb{Z}/15\mathbb{Z})$.

(1) **Case 1600.i1** There are five index 5 subgroups. Each of them contains a matrix of the form $\left[\begin{smallmatrix} 2 & 3k+2 \\ 0 & 1 \end{smallmatrix}\right]$, and all of them contain $\left[\begin{smallmatrix} 11 & 0 \\ 10 & 1 \end{smallmatrix}\right]$. They both have to be in $\rho_{E',15}(G_K)$ as their negatives do not have 1 as an eigenvalue mod 5. However, their product does not have 1 as an eigenvalue mod 3.

(2) **Case 1600.i2** Each of the index 5 subgroups contains a matrix of the form $\left[\begin{smallmatrix}2 & * \\ 0 & 11\end{smallmatrix}\right]$. Those matrices do not have eigenvalue 1 mod 3, and their negatives do not have eigenvalue 1 mod 5.
(3) **Case 1600.i3** Each of the index 5 subgroups contains a matrix of the form $\left[\begin{smallmatrix}6 & 3k+2 \\ 10 & 7\end{smallmatrix}\right]$ and the matrix $\left[\begin{smallmatrix}11 & 5 \\ 0 & 1\end{smallmatrix}\right]$, whose negatives do not have 1 as an eigenvalue mod 5, and whose product does not have 1 as an eigenvalue mod 3.
(4) **Case 1600.i4** The matrix $\left[\begin{smallmatrix}11 & 0 \\ 0 & 11\end{smallmatrix}\right]$ is in all index 5 subgroups.

We can use the same reasoning for points of order 21. We just need to consider elliptic curves in the isogeny class 1296.f. There are four such curves. In this case, there are no index 5 subgroups, so we need to look at index 7 subgroups.

For each of the four curves in the isogeny class, the mod 21 image has seven subgroups of index 7. Each of those contains an element which does not have neither 1 nor -1 as an eigenvalue modulo 7, which completes the proof. □

4.3. **Combining the 2^k-torsion with the odd part.** Note that for any $n > 2$, the entire n-torsion $E[n]$ cannot be defined over an odd degree number field K, as $K(\zeta_n)$ is contained in the field $K(E[n])$ and ζ_n is not contained in K.

As there are no points of order 16 this means that the possible 2^k-torsion over K is either trivial or of the form $\mathbb{Z}/2^k\mathbb{Z}$ or $\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2^k\mathbb{Z}$ for $k \in \{1, 2, 3\}$.

It remains to combine possible 2-torsion with the possible groups of points of odd order.

Lemma 4.8. *There are no $\mathbb{Q}$-curves over number fields of degree 5 or 7 with a point of order 14.*

Proof. If E was such a $\mathbb{Q}$-curve, then there would be a rational elliptic curve E'' with a 14-isogeny over an extension of degree 5 or 7, which is impossible by [16, Theorem 1.3], unless E'', and hence also E, has CM. However, by [3, 5.5 and 5.7], this is also not possible. □

Lemma 4.9. *There are no $\mathbb{Q}$-curves over number fields of degree 5 or 7 with a point of order 18 or 24, or independent points of order 2 and order 12.*

Proof. If E is a $\mathbb{Q}$-curve over a number field of degree 5 or 7 with the mentioned torsion group, we can assume that there is a $2^k \cdot 3^\ell$-isogeny to an elliptic curve E' with rational j-invariant. Let $E''/\mathbb{Q}$ be the rational twist of E', and let E''' be the corresponding twist of E. Then there is a 2^k3^ℓ-isogeny $E'' \to E'''$ defined over K. Since 5 and 7 do not divide $\#\mathrm{GL}_2(\mathbb{Z}/3\mathbb{Z})|$ and $\#\mathrm{GL}_2(\mathbb{Z}/2\mathbb{Z})$, it follows that this isogeny is in fact defined over $\mathbb{Q}$, so E''' is also rational. We conclude that E has rational j-invariant. By the results in [6] for K of degree 5, and [15] for K of degree 7, and using [9, Theorem 5.1.2], we reach a contradiction. □

Lemma 4.10. *There are no $\mathbb{Q}$-curves over number fields of degree 7 with a point of order 20 or independent points of order 2 and 10.*

Proof. We apply the same argument as in the proof of the previous lemma but with an isogeny of degree 2^k5^ℓ, noting that 7 does not divide $\#\mathrm{GL}_2(\mathbb{Z}/5\mathbb{Z})$ and $\#\mathrm{GL}_2(\mathbb{Z}/2\mathbb{Z})$. We obtain an elliptic curve with rational j-invariant and it cannot have a point of order 20 or independent points of order 2 and 10 by [9, Theorem 5.1.2]. □

The proof of Theorem 1.11 is now complete, i.e. the possible torsion subgroups of elliptic curves over septic fields have all been determined..

Lemma 4.11. *There are no $\mathbb{Q}$-curves over number fields of degree 5 with independent points of order 2 and 10, or a point of order 20.*

Proof. Let E be a $\mathbb{Q}$-curve over a number field K of degree 5 with two independent points of order 2 and 10 or with a point of order 20. In both of these cases, E is K-isogenous to an elliptic curve E_1 with a 20-isogeny. The elliptic curve E_1 is $2^a \cdot 5^b$-isogenous to an elliptic curve E' with rational j-invariant. We may factor this $2^a \cdot 5^b$-isogeny as a 5^b-isogeny $E \to E_2$ and a 2^a-isogeny $E_2 \to E'$. Since 5 does not divide $\#\mathrm{GL}_2(\mathbb{Z}/2\mathbb{Z})$, it follows that E_2 also has rational j-invariant, so we may assume that $a = 0$ and that E_1 is 5^b-isogenous to E'.

However, E' has a 4-isogeny over K since E_1 has a 4-isogeny. Since it also has a 5-isogeny, it has a 20-isogeny. But 20 is not a possible isogeny degree of an elliptic curve with rational j-invariant, so we have reached a contradiction. □

Lemma 4.12. *There are no $\mathbb{Q}$-curves over number fields of degree 5 with a point of order 50.*

Proof. Let E be a $\mathbb{Q}$-curve over a quintic number field K with a point P of order 50. Then there is an isogenous elliptic curve E' with rational j-invariant. As in the previous proof, we may assume that the isogeny $E \to E'$ is a 5^b-isogeny for some $b \geq 1$. If $b \geq 3$, then E' is an elliptic curve with rational j-invariant and a 250-isogeny over a quintic extension, which is impossible by [16, Theorem 1.3]. If the isogeny $E \to E'$ does not factor through the 50-isogeny whose kernel is the point of order 50, then E' again has a $2 \cdot 5^{2+b}$-isogeny which is impossible. Thus, $b \leq 2$ and the isogeny $E \to E'$ factors through the isogeny whose kernel is the point of order 50.

Suppose that $b = 2$. We have the following diagram.

$$E \text{ —— } \star \text{ —— } E' \text{ —— } \bullet$$

By Theorem 2.8, we know that some quadratic twist E'' of E' has a rational 5-isogeny, so either the elliptic curve marked with $\star$ or the elliptic curve marked with $\bullet$ has rational j-invariant. If the elliptic curve marked with $\bullet$ has rational j-invariant, then it has a 250-isogeny over K, a contradiction. Thus, the elliptic curve marked with $\star$ has rational j-invariant. Thus we may actually assume that $b = 1$ and treat the elliptic curve marked with $\star$ as E'.

By Theorem 2.5, $\rho_{E',50}(G_K)$ is contained in the subgroup of the group of all matrices of the form $\left[\begin{smallmatrix} 1 & 5\cdot k \\ 10\cdot j & z \end{smallmatrix}\right]$. It follows that $\#\rho_{E',25}(G_K)$ divides $5^3 \cdot 4$. Let E'' be a quadratic twist of E' which is defined over $\mathbb{Q}$. Then $\rho_{E'',25}(G_K)$ has cardinality dividing $5^4 \cdot 8$. Also, E'' has a rational 5-isogeny by Theorem 2.8. By [8, Theorem 2], if none of the curves in the $\mathbb{Q}$-isogeny class of E'' has two independent 5-isogenies, then $\rho_{E'',5^\infty}(G_\mathbb{Q})$ contains a Sylow pro-p subgroup. However, the Sylow pro-p subgroup when viewed modulo 25 has 5^5 elements, which does not divide $5^4 \cdot 8$. Thus, one of the curves in the $\mathbb{Q}$-isogeny class of E'' has two independent rational 5-isogenies. Since E'' also has a rational 2-isogeny coming from E, it follows that the isogeny class a rational elliptic curve with a 50-isogeny, which is a contradiction. □

This completes the proof of Theorem 1.10, i.e. the possible torsion subgroups over quintic fields have all been determined.

5. Quadratic torsion of elliptic curves isogenous to those with rational j-invariant

In this section, we prove Theorem 1.14.

Proposition 5.1. *Let E be an elliptic curve over a quadratic number field K which is $\overline{\mathbb{Q}}$-isogenous to an elliptic curve with rational j-invariant. Then $E(K)_{tors}$ is not isomorphic to $\mathbb{Z}/14\mathbb{Z}$ or $\mathbb{Z}/18\mathbb{Z}$.*

Proof. Suppose that $E(K)$ has torsion $\mathbb{Z}/14\mathbb{Z}$. Let E' be the elliptic curve with rational j-invariant isogenous to E. Applying a quadratic twist, one may assume that E' itself is defined over K. Then, by [4, Lemma A.4], there is a quadratic twist E'' of E' such that there is an isogeny $E \to E''$ defined over K itself. By Theorem 2.5, it follows that E'' has a 14-isogeny over K, which is not a possible isogeny degree of elliptic curves with rational j-invariant over quadratic fields by [19].

Suppose that $E(K)$ has torsion $\mathbb{Z}/18\mathbb{Z}$. By the results in [1, Section 4.6.], it follows that E is 2-isogenous to its Galois conjugate. If E is isogenous to an elliptic curve with rational j-invariant then, using notation from [4, Appendix], the j-invariant of E belongs to a rational class $\mathcal{Q}$. However, the degree of an isogeny from E to its conjugate then has to be a square by [4, Corollary A.12] and it cannot be 2. □

Proof of Theorem 1.14. For $p > 2$ this follows from Theorem 1.13, as elliptic curves over odd prime degree number fields K are exactly those isogenous to elliptic curves with rational j-invariant.

For $p = 2$, we know from the list of all possible torsion subgroups over quadratic fields by the results from Kamienny [10], and Kenku and Momose [11]. Najman [15, Theorem 2] found the list of all torsion subgroups of rational elliptic curves over quadratic extensions, and Gužvić [9, Theorem 5.1.2.] found the list of all torsion subgroups of elliptic curves with rational j-invariant over quadratic extensions.

It follows from these results that $\mathbb{Z}/14\mathbb{Z}$ and $\mathbb{Z}/18\mathbb{Z}$ are the only torsion subgroups that appear for some elliptic curves over quadratic fields and do not appear for elliptic curves with rational j-invariant. By Theorem 5.1, they also do not appear for elliptic curves isogenous to those with rational j-invariant, which completes the proof. □

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